

STA 130A: Mathematical Statistics: Brief Course

Lecture 5: Counting and Discrete Random Variables

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Agenda

Last time:

- Independence and independent trials
- The counting principle

Today:

- Permutations and combinations
- Discrete random variables and probability mass functions
- Common discrete models: Bernoulli, binomial, geometric, and Poisson

Recap: The counting principle

For a finite sample space with equally likely outcomes,

$$P(A) = \frac{|A|}{|\Omega|}.$$

Thus many probability problems reduce to counting

The counting principle supports a divide-and-conquer approach

The counting principle: multiply the number of choices across stages

- The number of choices at each stage must be the same for every earlier history
- Three binary stages give $2 \times 2 \times 2 = 8$ sequences

Today: How can we systematically count when **order matters** or **does not matter**?

Permutations

A **k -permutation** of n distinct objects is an *ordered* selection of k of them

- The **factorial** of a nonnegative integer:

$$n! = n(n-1) \cdots 2 \cdot 1, \quad \text{with convention} \quad 0! := 1$$

- The number of **k -permutations** is

$$n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}$$

- As a special case, the number of permutations of n distinct objects is

$$n!$$

- **Example:** Choosing 2 objects from 4 in order gives

$$4 \cdot 3 = \frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1} = 12$$

Permutations: Examples

Example (Word of four distinct letters)

Question: How many 4-letter strings can be formed using distinct letters from the alphabet (A to Z)?

Answer: The total number of possible words is

$$\frac{n!}{(n-k)!} = \frac{26!}{22!} = 26 \cdot 25 \cdot 24 \cdot 23 = 358,800$$

Example (CDs of three types)

Suppose you have n_1 classical CDs, n_2 rock CDs, and n_3 country CDs (all distinct)

Question: In how many different ways can you arrange them so that CDs of the same type are contiguous?

Answer: Count in two stages:

- Choose the order of the three blocks: $3!$ ways
- Arrange the CDs within each block: $n_1!$, $n_2!$, and $n_3!$ ways

Therefore,

$$\#\{\text{valid arrangements}\} = 3! n_1! n_2! n_3!$$

Combinations

A **combination** is a selection in which order does *not* matter

- A **k -combination** of n distinct objects is a k -element subset
- Start from ordered selections (= permutations) and group together duplicate orderings
 - Divide by the $k!$ reorderings of the same subset
 - The number of possible combinations is equal to the **binomial coefficient**

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- **Example:** The number of distinct 2-subsets of a 4-element set $\{A, B, C, D\}$ is

$$\binom{4}{2} = 6$$

- The 2-element subsets of $\{A, B, C, D\}$ are

AB, AC, AD, BC, BD, CD

Combinations: Example

Example (Exactly k Heads)

Question: In n independent Bernoulli trials, how many outcomes contain exactly k Heads?

Key observaiton: A sequence with exactly k Heads is determined by the k positions of the Heads

Therefore,

$$\#\{\text{outcome sequences with exactly } k \text{ Heads}\} = \binom{n}{k}$$

- For example, with $n = 3$ and $k = 2$, there are $\binom{3}{2} = 3$ sequences:

HHT, HTH, THH

This counts possible sequences only; probabilities will also depend on the trial model

If each trial has the same Head probability $p \in [0, 1]$, then

$$P(k \text{ Heads}) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Pop-up quiz

Question: How many different ways can we select a president and a vice president from 6 students?

a) $6 + 5$

b) $6 \cdot 5$

c) $6!$

d) $\binom{6}{2} := \frac{6!}{2!(6-2)!}$

Answer: B. There are 6 choices for president and then 5 remaining choices for vice president, so the answer is

$$6 \cdot 5 = 30$$

Follow-up: How many different choices of 2-person committees are there?
(Note: the order does not matter here)

Motivation: From outcomes to numbers

In many probabilistic models, the outcomes are numerical

- Sometimes outcomes themselves are numeric
- Even if not, we care about some numbers related to outcomes

A random variable X extracts a numerical feature of the outcome

- 3 coin tosses $\rightarrow$ number of Heads
- 2 die rolls $\rightarrow$ sum of the faces
- game outcome $\rightarrow$ monetary payoff

Once outcomes are mapped to numbers, we can ask questions such as

- What is the probability of $X = 2$?
- What is the probability of $X \geq 3$?
- What is the "most likely" value of X ?

Random variable: Definition

Definition (Random variable)

A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$

- A random variable assigns a real number to each outcome $\omega \in \Omega$
- The randomness comes from the outcome ω , not from the function X itself
- X is **discrete** if its range is finite or countably infinite

Examples:

- *Discrete*: number of Heads in n coin tosses; number of trials until the first Head
- *Continuous*: landing location of a dart on a line segment

The event $\{X = x\}$ is a subset of $\Omega \rightarrow$ questions about X are still questions about events

Notation: For simplicity, we may write

$$P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\}), \quad P(X \in S) = P(\{\omega \in \Omega : X(\omega) \in S\})$$

Probability mass function

Definition (Probability mass function)

If X is a discrete random variable, its **probability mass function (PMF)** is a function $p_X : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$p_X(x) = P(X = x), \quad x \in \mathbb{R}$$

The **support** of X is

$$\mathcal{X} = \{x \in \mathbb{R} : p_X(x) > 0\}$$

- $p_X(x) \geq 0$ for all x
- $\sum_{x \in \mathcal{X}} p_X(x) = 1$, where $\mathcal{X} = \{x : p_X(x) > 0\}$
- For any set $S \subseteq \mathbb{R}$,

$$P(X \in S) = \sum_{x \in S \cap \mathcal{X}} p_X(x)$$

The PMF completely determines the distribution of a discrete random variable

Probability mass function: Example

Example (2 independent fair coin tosses)

Sample space: $\Omega = \{HH, HT, TH, TT\}$

Define a random variable X denoting the number of Heads:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

(For example, the event $\{X = 1\}$ is the set $\{HT, TH\}$)

The support is $\{0, 1, 2\}$, and because the four outcomes are equally likely,

$$P(X = 2) = 1/4, \quad P(X = 1) = 1/2, \quad P(X = 0) = 1/4$$

Therefore, the PMF of X is

$$p_X(x) = \begin{cases} 1/4, & x = 0, \\ 1/2, & x = 1, \\ 1/4, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Pop-up quiz

For two independent fair coin tosses, let X be the number of Heads.

Question: Which statement is correct?

- A) $p_X(HT) = 1/4$
- B) $\{X = 1\} = \{1\}$
- C) $P(X \geq 1) = p_X(1) + p_X(2)$
- D) The support of X is $\{HH, HT, TH, TT\}$

Answer: C.

Because $X \geq 1$ means $X = 1$ or $X = 2$, so we sum the PMF values at 1 and 2. Also, $\{X = 1\} = \{HT, TH\}$, and the support is $\{0, 1, 2\}$.

Follow-up: What is $P(X \text{ is even})$?

Bernoulli random variable

Purpose: One success/failure trial

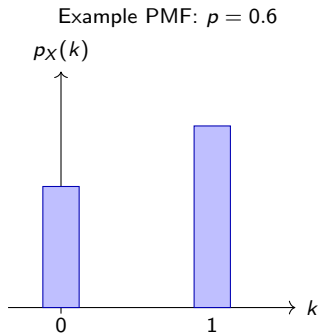
Notation: For $p \in [0, 1]$,

$$X \sim \text{Bernoulli}(p)$$

PMF:

$$p_X(k) = \begin{cases} p, & k = 1, \\ 1 - p, & k = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Support: $\{0, 1\}$



- **Interpretation:** $X = 1$ means success, $X = 0$ means failure
- **Example:** One coin toss; whether an email is spam
- Bernoulli variables are the building blocks for many common probabilistic models

Binomial random variable

Purpose: Number of successes in n independent Bernoulli(p) trials

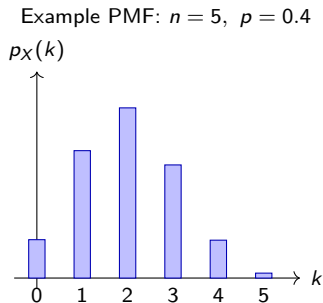
Notation: For $n \in \{1, 2, \dots\}$ and $p \in [0, 1]$,

$$X \sim \text{Binomial}(n, p)$$

PMF:

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Support: $\{0, 1, \dots, n\}$



- **Interpretation:** $X = k$ means exactly k successes occur in n trials
- **Example:** Number of Heads in n coin tosses
- $X = X_1 + \dots + X_n$, where $X_1, \dots, X_n$ are independent Bernoulli(p)

Geometric random variable

Purpose: Waiting time until the first success in repeated independent Bernoulli(p) trials

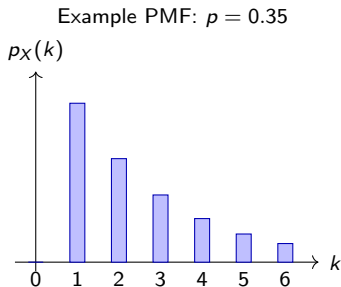
Notation: For $p \in [0, 1]$,

$$X \sim \text{Geometric}(p).$$

PMF:

$$p_X(k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

Support: $\{1, 2, 3, \dots\}$



- **Interpretation:** $X = k$ means the first success occurs on trial k
- **Example:** Number of tosses until the first Head
- The trials are i.i.d. Bernoulli(p); we stop at the first success

Poisson random variable

Purpose: Number of occurrences in a fixed time or space window

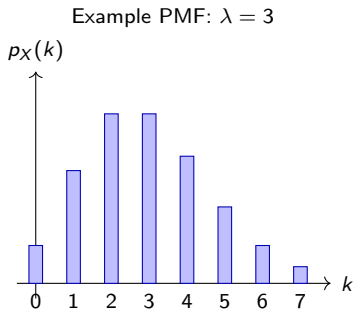
Notation:

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0.$$

PMF:

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Support: $\{0, 1, 2, \dots\}$



- **Interpretation:** $X = k$ means the event occurs k times in the window
- **Example:** Number of emails in an hour; number of typos on a page
- Poisson RV can arise as a limit of many rare Bernoulli opportunities with $np \approx \lambda$

Pop-up quiz

Suppose independent Bernoulli(p) trials are repeated until the first success occurs

Question: Which model best describes the random variable

X = the trial on which the first success occurs?

- A) Bernoulli(p)
- B) Binomial(n, p)
- C) Geometric(p)
- D) Poisson(λ)

Answer: C.

Because X is a waiting time to the first success in repeated independent Bernoulli trials.

Follow-up: If we instead run exactly n trials and count the total number of successes, which model do we get?

Wrap-up

Random variables

outcomes $\omega \in \Omega \longrightarrow X(\omega) \in \mathbb{R} \longrightarrow p_X(x) = P(X = x)$

- A random variable is a function $X : \Omega \rightarrow \mathbb{R}$ that assigns a number to each outcome
- Discrete random variables take finitely many or countably many values

Probability mass functions

- The PMF is $p_X(x) = P(X = x)$
- For a discrete random variable, $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$
- For any set $S \subseteq \mathbb{R}$, $P(X \in S) = \sum_{x \in S \cap \mathcal{X}} p_X(x)$

Common discrete models

- Bernoulli: one success/failure trial
- Binomial: number of successes in n independent Bernoulli trials
- Geometric: number of trials until the first success
- Poisson: number of occurrences in a fixed window

Suggested reading: [BT08, Ch. 1.6 & 2.1–2.2]

References



Dimitri Bertsekas and John N Tsitsiklis.

Introduction to probability, volume 1.

Athena Scientific, 2nd edition, 2008.